

1. Sample sizes

Eos in APOGEE + LAMOST - selection & orbits (generated 2026-06-19)

APOGEE (ASPCAP allStar chemistry + AstroNN orbits, row-matched)

```
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base sample (clean red giants, SNR>70, no dups, dist<15 kpc) : 274,175
Eos : 147
...of which with reliable age (rel.err < 20%) : 4
Splash : 5,149
low-alpha disc (reference) : 180,976
```

LAMOST (DD-Payne abundances x subgiant MST0 ages, joined on SPECID)

```
-----
joined base sample (DR9 DD-Payne x subgiant_fullparam_update) : 244,105
Eos : 305
...of which with reliable MST0 age (rel.err < 20%) : 260
Splash : 2,994
low-alpha disc (reference) : 149,732
```

Notes

- * APOGEE giants give clean chemistry incl. [Al/Fe] but coarse ages that saturate above ~10 Gyr, so only 4 Eos stars have reliable APOGEE ages.
- * LAMOST subgiants give precise MST0 ages; [Al/Fe] is supplied by the DR9 DD-Payne catalogue (median AL_FE_ERR ~ 0.15 dex).
- * The two surveys overlap in only ~6 stars by Gaia source_id (giants vs subgiants are largely disjoint), so each survey is selected internally; there is no star-by-star cross-survey transfer.

2. Exact cuts - APOGEE

Eos in APOGEE + LAMOST - selection & orbits (generated 2026-06-19)

Data sources (the paper's choice):

```
chemistry -> ASPCAP allStar (APOGEE_DR17_all.fits); MG_FE, AL_FE, FE_H
            already [X/Fe] / [Fe/H]. (AstroNN neural-net abundances
            are NOT used: they show a spurious gap at [Fe/H]~-0.55.)
orbits/age -> AstroNN VAC, row-matched by APOGEE_ID.
Kinematics: Vtan = galvt   ecc = e   rap = rap   Lz = Lz   E = Energy
```

(a) Base sample (clean of duplicates / unreliable measurements; red giants)

```
weighted_dist < 15 kpc
FE_H, MG_FE, AL_FE not fill (> -100) and finite
FE_H_FLAG = MG_FE_FLAG = AL_FE_FLAG = 0   (reliable measurements)
ASPCAPFLAG STAR_BAD (bit 23) not set; SNR > 70
red giants: 1.0 < LOGG < 3.5
duplicates removed: one row per APOGEE_ID, highest SNR
```

(b) Chemical sequence definitions

```
in-situ      : [Al/Fe] > -0.12
accreted     : [Al/Fe] <= -0.12
Mg split     : S(feh) = 0.16 - 0.08*[Fe/H]
high-alpha   : in-situ AND [Mg/Fe] > S(feh)
low-alpha    : in-situ AND [Mg/Fe] <= S(feh) AND [Fe/H] > -1.1
               AND ( [Fe/H] >= -0.6 OR [Al/Fe] > 0.9*[Fe/H] + 0.9 )
               [the diagonal cleans the metal-poor extension only]
```

(c) Kinematic operationalisation

```
halo orbit   : |Vtan| < 100 km/s AND ecc > 0.6
disc orbit   : Vtan > 150 km/s AND ecc < 0.35
```

(d) Final samples

```
Eos          : low-alpha AND halo-orbit AND -1.1 < [Fe/H] < -0.5
Splash       : high-alpha AND [Fe/H] > -0.9 AND |Vtan| < 80 km/s
low-a disc   : low-alpha AND disc-orbit
```

The high/low-alpha split slope/intercept and the halo-orbit thresholds are tuned to reproduce Fig. 1 of the draft; all live in `src/eos/config.py` and are applied in `src/eos/selections.py`. The $[Al/Fe] > 0.9[Fe/H] + 0.9$ diagonal is taken verbatim from the paper.

3. Exact cuts - LAMOST

Eos in APOGEE + LAMOST - selection & orbits (generated 2026-06-19)

Source catalogues (joined on LAMOST SPECID):

abundances : LMDR9_DDPAYNE_recommend_202505.fits (MG_FE, AL_FE, FEH)
-> already [X/Fe]; AL_FE used directly
ages+orbits: subgiant_fullparam_update.fits
AGE, AGE_ERR, ECC, R_APO, R_PERI, LZ, E, VT (=Vtan)

Identical selection logic to APOGEE (same constants), now with LAMOST cols:

in-situ : $AL_FE > -0.12$
Mg split : $S(feh) = 0.16 - 0.08*FEH$
low-alpha : $in-situ \text{ AND } MG_FE \leq S(feh) \text{ AND } FEH > -1.1$
 $\text{AND } (FEH \geq -0.6 \text{ OR } AL_FE > 0.9*FEH + 0.9)$
halo orbit : $|VT| < 100 \text{ km/s AND } ECC > 0.6$
disc orbit : $VT > 150 \text{ km/s AND } ECC < 0.35$

Eos : low-alpha AND halo-orbit AND $-1.1 < FEH < -0.5$
Splash : high-alpha AND $FEH > -0.9$ AND $|VT| < 80 \text{ km/s}$

Age figures additionally require relative age error < 20%.

Caveat specific to LAMOST: the DD-Payne aluminium error (~ 0.15 dex) is $\sim 3\times$ the APOGEE value, so the in-situ/accreted ($[Al/Fe] > -0.12$) boundary is noisier here; the metal-poor low-alpha selection leans mainly on $[Mg/Fe]$ and $[Fe/H]$, with $[Al/Fe]$ as a secondary discriminant.

4. How `r_apo` (and other orbital quantities) were computed

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No orbital integration was run in this project. All orbital quantities (`r_apo`, `r_peri`, eccentricity, `Lz`, Energy, `zmax`, actions) were read from PRE-COMPUTED columns in the source catalogues.

APOGEE (used for the orbit-vs-Splash comparison, Fig. ana3):

column : rap (also rperi, e, Lz, Energy, ...)
source : apogee_astroNN-DR17.fits (AstroNN VAC)
computed by : Mackereth & Bovy (2018); Leung & Bovy (2019)
integration code : galpy (Bovy 2015)
Galactic potential: MWPotential2014 (Bovy 2015)
6D input : Gaia EDR3 astrometry + AstroNN spectro-photometric distances + APOGEE line-of-sight velocities

LAMOST (used for the age analyses):

columns : R_APO, R_PERI, ECC, LZ, E (subgiant catalogue)
source : subgiant_fullparam_update.fits
computed by : the subgiant-catalogue authors (pre-computed),
from Gaia astrometry + LAMOST velocities + subgiant spectro-photometric distances.

Consistency note:

The quantitative `r_apo` / `ecc` / `Lz` comparison between Eos and the Splash (KS tests in results/README.md) was done WITHIN APOGEE only, so both samples share the same MWPotential2014 / galpy orbit pipeline and the comparison is internally consistent.

Unit conventions used in figures:

$E \times 10^{-5}$ (Energy in km^2/s^2) $Lz \times 10^{-3}$ (Lz in kpc km/s)
`r_apo`, `r_peri` in kpc ; `Vtan` in km/s .